that stabilize flight paths and make corrective decisions based on sensor inputs. Improved battery technology, particularly lithium-polymer (Li-Po) and lithium-ion systems, extended flight times and supported heavier payloads such as high-resolution cameras, multispectral sensors, and delivery packages.

The most transformative aspect of this generation is its expansion into commercial, industrial, and public-sector applications. In agriculture, drones became indispensable for precision farming, monitoring crop health, mapping fields, and applying fertilizers or pesticides with pinpoint accuracy. In construction and infrastructure, UAVs enabled aerial surveying, structural inspections, and 3D mapping. Environmental monitoring, disaster management, and public safety operations leveraged drones for real-time situational awareness, search and rescue, and post-disaster damage assessment. The logistics and delivery sector also began exploring UAV integration for lastmile transport, with prototypes successfully demonstrating automated package delivery.

From a regulatory perspective, this generation coincided with the establishment of civil aviation frameworks by organizations such as the FAA (Federal Aviation Administration), EASA (European Union Aviation Safety Agency), and DGCA (Directorate General of Civil Aviation, India). These frameworks introduced airspace classifications, pilot certification, and safety guidelines that allowed drones to be integrated safely into national airspaces.

The fourth-generation drones symbolize the convergence of autonomy, digitalization, and industry-focused innovation. They bridge the gap between fully manual systems and next-generation intelligent UAVs by enabling adaptive control, autonomous missions, and vast commercial scalability. This generation positioned drones as indispensable tools across